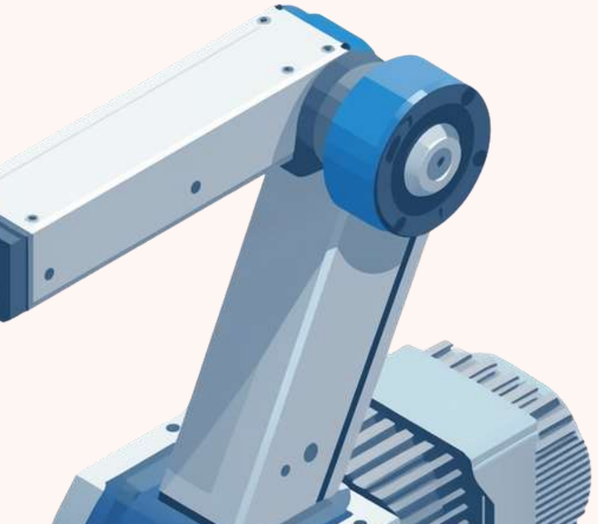
IR CONTROLLED ROBOTIC ARM



RoboForge Project

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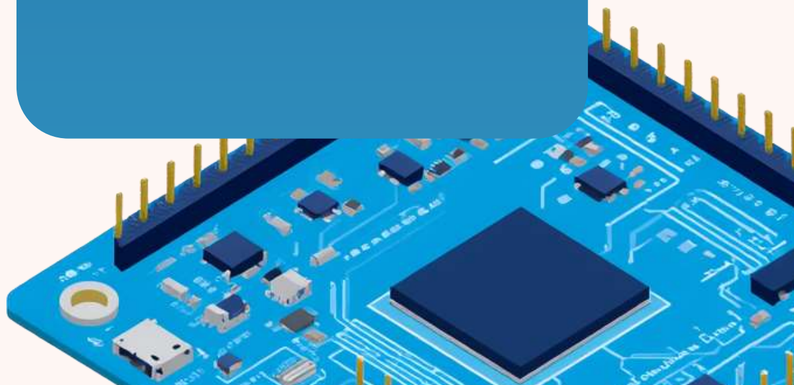
PROJECT OVERVIEW

Control System
- IR remote controls arm movement directions
- Potentiometer for precise base rotation
- Real-time responsive input handling
- Intuitive user interaction design

Core Platform
- Built on RP2040 (Shrike board)
- Powerful microcontroller processing
- Real-time movement control
- Designed for learning robotics basics

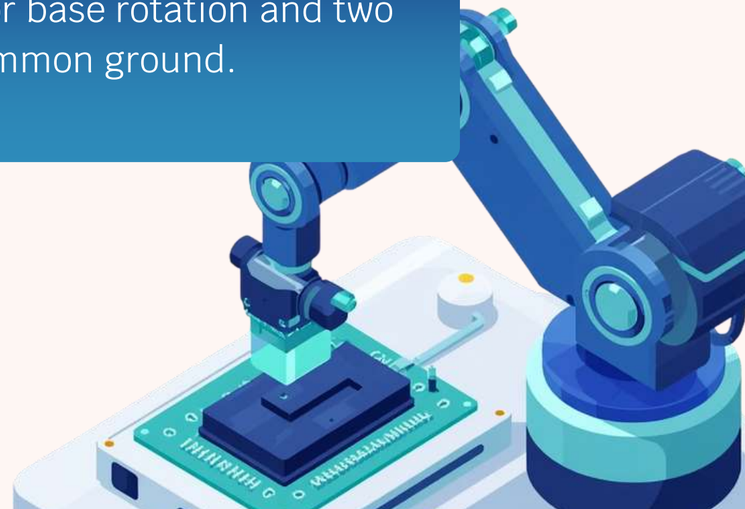
Hardware Components
- MG996R servo (base rotation)
- 2 × SG90 servos (arm joints)
- IR sensor + remote module
- Potentiometer for analog input
- External 5V power supply



SYSTEM WORKING

Input Stage: The potentiometer provides analog voltage signals to control base rotation angle to MG996R Base servo . The IR remote transmits infrared commands detected by the IR sensor for arm movement control. Both inputs feed into the RP2040 microcontroller for real-time processing.

Processing & Output: The RP2040 reads analog signals from the potentiometer and decodes IR commands, converting these inputs into precise servo angles. Output drives the MG996R servo for base rotation and two SG90 servos for arm movement. External 5V supply ensures stable operation with common ground.



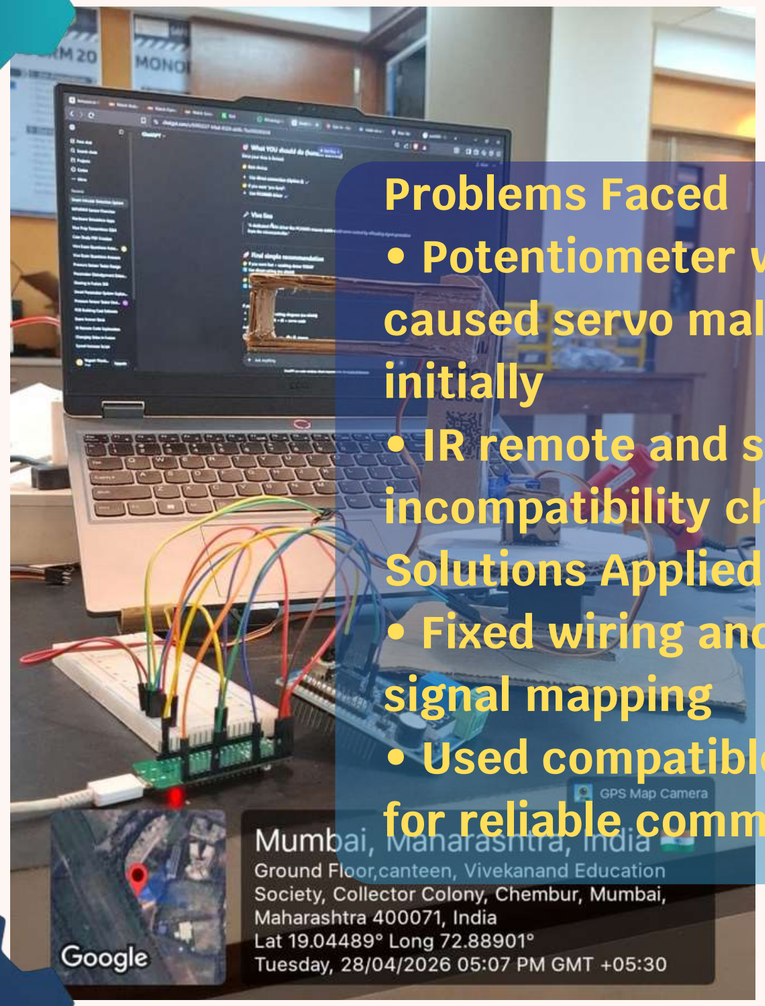
CODE LOGIC

Software Operation Steps:
- System initializes servos and IR receiver at startup
- Continuously reads potentiometer for base rotation control
- Detects IR commands for arm movement control
- Uses PWM signals to achieve smooth servo motions

Safety Mechanisms:
- Stops servo movement when no signal is received
- Angle limits enforced using constrain() function
- Prevents erratic behavior and protects hardware

Flow Summary:
Start → Initialize → Read Inputs → Process → Move Servos → Repeat
This continuous loop ensures real-time responsive control of all servo motors.

CHALLENGES & LEARNING

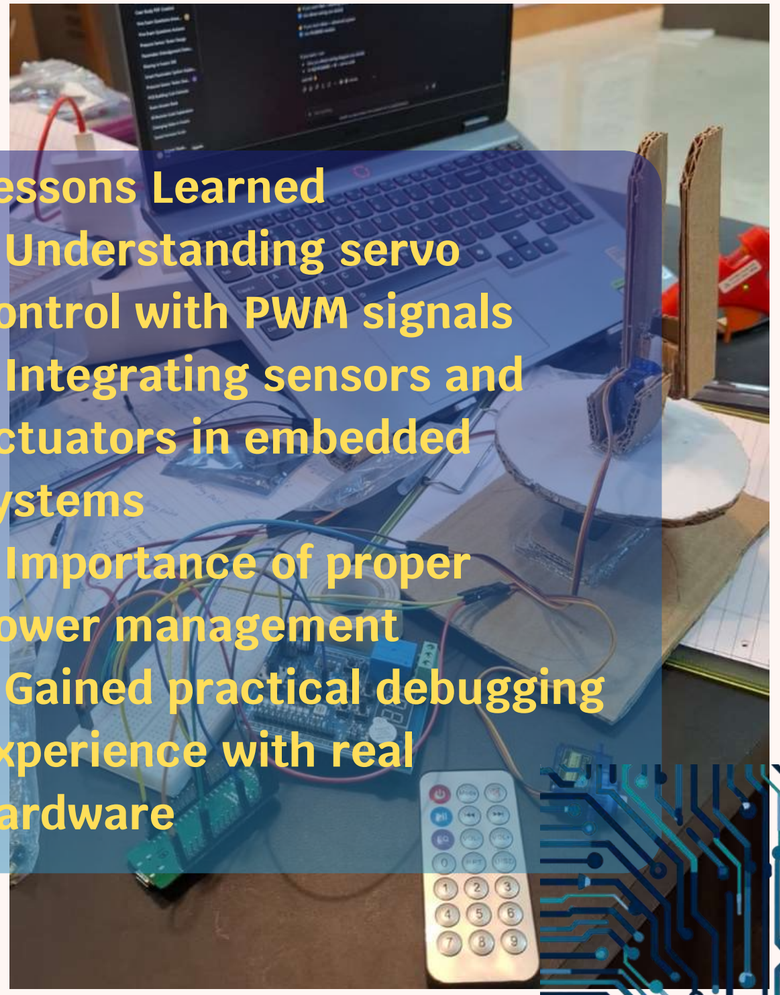


Problems Faced

- Potentiometer wiring issue caused servo malfunction initially
- IR remote and sensor incompatibility challenges

Solutions Applied

- Fixed wiring and corrected signal mapping
- Used compatible IR codes for reliable communication



Lessons Learned

- Understanding servo control with PWM signals
- Integrating sensors and actuators in embedded systems
- Importance of proper power management
- Gained practical debugging experience with real hardware